Self-solutions to Infra-Bayesianism Problem Set Solutions authored by Diffractor et. al *

ZKMathquant

Source:ZKMathquant/ Solving-IB-and-coding-RL-agents-getting-hands-on/tree/main/ibmath

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*Original problem set authored by Connall Garrod, Jack Parker, and Diffractor.,found in this post here

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1. Assignment- I

Unless specified otherwise,most of the content here is borrowed from this Assignment document found in
this alignment forum post "infra-exercises-part-1".

The current work is only to add self-thought solutions to the problems, and anywhere else lacking a solid proof or interesting enough to brainstorm own proofs.

The part,mostly proofs, contributed by the author here will be colored like this

1.1. Section-I

Definition 1.1 (Mixing points and mixture)**.** Let $x, x' \in X$, and let $p \in [0, 1]$. The mixture (also called the affine combination) of x and x' is the value

$$px + (1 - p)x'$$

A mixture is just a weighted average of points (you can mix more than two points). In the case where $p = 0.5$, the mixture of two points is just the arithmetic mean. If you want "a little more x" in your mixture, just dial up the value of p (called the mixing coefficient), and vice versa.

Definition 1.2 (Affine function)**.** Let X and Y be spaces in which one can mix points. We say that $f : X \to Y$ is affine if and only if for all $x, x' \in X$ and $p \in [0, 1]$,

$$f(px + (1 - p)x') = pf(x) + (1 - p)f(x')$$

In other words, mixing inputs before applying f yields the same result as applying f to each input and then mixing the outputs.

Exercise-1.1

> Have a think about whether the following functions are affine: parabola, line through the origin, piecewise linear function, line not through the origin. Sketch graphs to help yourself visualize what's going on.

Turns out from graphs only line through or not-through the origin is affine, not even the piecewise linear function.

Definition 1.3 (Linear function)**.** Let V and W be vector spaces over a field of scalars F (for simplicity, you can think of F as $\mathbb{R}$). A function $f : V \to W$ is linear if and only if for all $\lambda \in F$ and for all vectors $x, x' \in V$:
1) $f(\lambda x) = \lambda f(x)$, and
2) $f(x + x') = f(x) + f(x')$.

Notice that the first condition guarantees that linear functions map the zero vector in V to the zero vector in W.

Problem 1 Let V be a vector space, and let $f : V \to \mathbb{R}$. Prove that f is affine if and only if $f(x) = g(x) + c$, where g is linear and $c \in \mathbb{R}$.

In the case where $f : \mathbb{R} \to \mathbb{R}$, the claim is that an affine function is a line that may or may not go through the origin.

Later on in this problem set, we will need to make use of the fact that affine functions can be split into a linear part and a constant part.

`Author's solution-1`

Proof. 1. $\implies$ part :

Define, $g(x) = f(x) - f(0)$. Enough to show,g is linear. Since, f is affine,we have, $\forall x, y, t$

$$f(tx + (1 - t)y) = tf(x) + (1 - t)f(y)$$

Put,y=0. We get,$\forall x, t,\ f(tx) = tf(x) + (1 - t)f(0) \implies f(tx) - f(0) = tf(x) - tf(0)$
$\implies g(tx) = tg(x)$

Thus,g is linear if we show $g(x+y) = g(x) + g(y)\ \forall x, y$.

For arbitrary x, y, Consider 0.5 mixture of x and y on the affinity of f:

$f(0.5(x+y)) = 0.5(f(x) + f(y))$
$\implies g(0.5(x+y)) + f(0) = 0.5(g(x) + g(y) + 2f(0))$
$\implies 0.5g(x+y) + f(0) = 0.5(g(x) + g(y)) + f(0)$
$\implies g(x+y) = g(x) + g(y)$

And,we are done

2. $\Leftarrow$ part:

   Assume,$\exists$ linear map $g \ni f(x) = g(x) + c\ \ \forall x \in V$ Therefore,$g(0) = 0 \implies f(0) = c \implies g(x) = f(x) - f(0)$

   We want to show,f is affine.

   Take arbitrary $x, y, \in V$ and an arbitrary $t \in [0, 1]$ mixture of x and y applied on f :

   $f(tx + (1-t)y) = g(tx + (1-t)y) + f(0) = g(tx) + g((1-t)y) + f(0)$[by linearity of g]

   $= tg(x) + (1-t)g(y) + f(0)$[again,by linearity of g] $= tf(x) - tf(0) + (1-t)f(x) - (1-t)f(0) + f(0)$
   $= tf(x) + (1-t)f(x)$

   ,i.e.,the desired mixture on the corresponding functional values of proving that f is affine.

□

1.2. Section-II

Definition 1.4 (Concavity)**.** Let X and Y be spaces in which one can mix points, and let Y be partially ordered (definition in appendix). We say that $f : X \to Y$ is concave if and only if for all $x, x' \in X$,

$$f(px + (1-p)x') \geq pf(x) + (1-p)f(x').$$

Geometrically, $f : \mathbb{R} \to \mathbb{R}$ is concave if and only if for any point z between x and x', $f(z)$ lies above or on the line segment joining $(x, f(x))$ and $(x', f(x'))$.

Sketch a graph to help yourself visualize this,for example,a parabola $y = x^2$ isn't concave.

Definition 1.5 (Function ordering)**.** Let Y be a partially ordered set, and let $f, g : X \to Y$. We say that $f \geq g$ if and only if $f(x) \geq g(x)$ for all $x \in X$. Note that this is a partial ordering of functions, because given f and g, there is no guarantee that they can be compared.

Definition 1.6 (Infimum of a set of real numbers)**.** Let $S \subset \mathbb{R}$. The infimum of S, denoted $\inf S$, is the greatest lower bound of S. More formally, m is the infimum of S if it has the following two properties:

1) For all $s \in S$, $m \leq s$;
2) For all $\varepsilon > 0$, there exists $s \in S$ such that $m < s < m + \varepsilon$.

Note that not all sets have an infimum, but we won't have to worry about that

Definition 1.7 (Infimum of a collection of functions)**.** Let $\{f_i\}_{i \in I}$ be a collection of functions of type signature $X \to Y$. Then the infimum of the collection of functions is the "greatest lower bound function" constructed by taking the pointwise infimum. In other words, for each $x \in X$, the value of the infimum at x is $\inf_{i \in I} f_i(x)$.

Definition 1.8 (Linear funcitonal)**.** A linear functional is a linear mapping from a vector space V to its field of scalars F(which is itself a vector space, trivially, to hold the Definition 1.3 of linear maps)

Theorem 1.9 (Hahn-Banach Separation Theorem)**.** *[Rudin 1991]*

Let V be a vector space, and let $C \subset V$ be closed and convex . Let $v \in V$ but $v \notin C$. Then there exist a continuous linear functional $F : V \to \mathbb{R}$ and $s \in \mathbb{R}$ such that

$$\forall c \in C : F(c) \leq s < F(v).$$

Proof. `A:Strong Separation in a Real Normed Space`

Let V be a real normed vector space, $C \subset V$ a non-empty closed and convex set, and $v \in V$ such that $v \notin C$. Then there exists a continuous linear functional $F : V \to \mathbb{R}$ and $s \in \mathbb{R}$ such that $F(c) \leq s < F(v)$ for all $c \in C$.

Step 1: Shift to the Origin. Define the set $C' = C - v = \{c - v : c \in C\}$. Since translation is a homeomorphism, C' is closed and convex. Because $v \notin C$, we have $0 \notin C'$.

Step 2: Utilize the Distance Property. In a normed space, the distance from a point to a closed set is positive. Define:

$$d = \inf_{x \in C'} \|x\|$$

Since C' is closed and $0 \notin C'$, it follows that $d > 0$. Consider the open ball $B = B(0, d) = \{x \in V : \|x\| < d\}$. By definition of d, $B \cap C' = \emptyset$.

Lemma 1.10 (Hahn-Banach Extension Theorem)**.** *Let V be a real vector space and $p : V \to \mathbb{R}$ be a sublinear functional (i.e., $p(x + y) \leq p(x) + p(y)$ and $p(\alpha x) = \alpha p(x)$ for $\alpha \geq 0$). Let M be a linear subspace of V and $f : M \to \mathbb{R}$ be a linear functional such that:*

$$f(x) \leq p(x) \quad \forall x \in M$$

Then there exists a linear functional $F : V \to \mathbb{R}$ such that $F|_M = f$ and:

$$F(x) \leq p(x) \quad \forall x \in V$$

Proof. **1. The One-Dimensional Extension Step:**
Suppose $M \neq V$ and pick $v_0 \in V \setminus M$. We wish to extend f to $M_1 = \text{span}\{M, v_0\}$ by defining $F_1(m + tv_0) = f(m) + t\alpha$ for some $\alpha \in \mathbb{R}$. For $F_1 \leq p$ to hold, we require:

$$f(m) + t\alpha \leq p(m + tv_0) \quad \forall m \in M, t \in \mathbb{R}$$

Analyzing $t > 0$ and $t < 0$ (letting $t = -k$ for $k > 0$):

1. $f(m/t) + \alpha \leq p(m/t + v_0) \implies \alpha \leq p(m_2 + v_0) - f(m_2)$
2. $f(m/k) - \alpha \leq p(m/k - v_0) \implies \alpha \geq f(m_1) - p(m_1 - v_0)$

Such an α exists if $\sup_{m_1 \in M}\{f(m_1) - p(m_1 - v_0)\} \leq \inf_{m_2 \in M}\{p(m_2 + v_0) - f(m_2)\}$. This is guaranteed by the sublinearity of p:

$$f(m_1) + f(m_2) = f(m_1 + m_2) \leq p(m_1 + m_2) = p(m_1 - v_0 + m_2 + v_0) \leq p(m_1 - v_0) + p(m_2 + v_0)$$

Rearranging gives $f(m_1) - p(m_1 - v_0) \leq p(m_2 + v_0) - f(m_2)$. Thus, α exists and f can be extended to M_1.

2. The Transfinite Extension (Zorn's Lemma):
Let $\mathcal{E}$ be the set of all pairs $(g, D(g))$ where $D(g)$ is a subspace such that $M \subseteq D(g) \subseteq V$, and g is a linear extension of f satisfying $g \leq p$ on $D(g)$. Order $\mathcal{E}$ by $(g_1, D(g_1)) \leq (g_2, D(g_2))$ if $D(g_1) \subseteq D(g_2)$ and $g_2|_{D(g_1)} = g_1$. Every chain in $\mathcal{E}$ has an upper bound (the union of the subspaces and functionals). By Zorn's Lemma, there exists a maximal element (F, V_{max}).

If $V_{max} \neq V$, we could apply the one-dimensional extension step to extend F further, contradicting maximality. Thus, $V_{max} = V$. □

Step 3: Apply the Minkowski Functional.
The ball B is a convex, open neighborhood of 0. Let p be the Minkowski functional of B.
$p_B(x) = \inf\{t > 0 : \frac{x}{t} \in B\}$
$p(x + y) \leq p(x) + p(y)$
$p(\lambda x) = \lambda p(x), \quad \forall \lambda \geq 0$
$p(x) = \frac{\|x\|}{d}$

Since B is the unit ball scaled by d, $p(x) = \frac{\|x\|}{d}$. By the 1.10,the Hahn-Banach Extension Theorem, there exists a linear functional F such that $F(x) \leq p(x)$ for all $x \in V$. Specifically:

1. For $x \in B$, $p(x) < 1$, so $F(x) < 1$. This implies F is continuous with $\|F\| \leq \frac{1}{d}$.
2. Since $B \cap C' = \emptyset$, for any $x \in C'$, $p(x) \geq 1$.

Step 4: Establish the Separation. Consider the functional $-F$. For any $c \in C$, let $x = c - v \in C'$. From the properties above, we can find a functional such that $F(c - v) \leq -1$. This implies:

$$F(c) - F(v) \leq -1 \implies F(c) \leq F(v) - 1$$

Let $s = F(v) - 1$. Then $F(c) \leq s < F(v)$, providing the strict separation.

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B:Strong Separation in a Locally Convex Space
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Let V be a real locally convex Hausdorff topological vector space, $C \subset V$ a closed convex set, and $v \notin C$. Then there exists a continuous linear functional $F \in V^*$ and $s \in \mathbb{R}$ such that $F(c) \leq s < F(v)$ for all $c \in C$.

Lemma 1.11. *Let X be a real locally convex TVS. Let $A \subset X$ be a non-empty open convex set, and $B \subset X$ be a non-empty convex set such that $A \cap B = \emptyset$.*

Then there exists a continuous linear functional $f \in X^*$ and a constant $\alpha \in \mathbb{R}$ such that:

$$f(a) < \alpha \leq f(b) \quad \forall a \in A, \forall b \in B$$

Proof. 1. Vector Subtraction and the Origin

Define the set $C = A - B = \{a - b : a \in A, b \in B\}$.

Convexity: Since A and B are convex, C is convex.

Openness: $C = \bigcup_{b \in B}(A - b)$. Since A is open, each translate $A - b$ is open. A union of open sets is open, so C is open.

Disjointness: Since $A \cap B = \emptyset$, the origin $0 \notin C$.

2. The Gauge Function (Minkowski Functional)

Fix some $x_0 \in C$. Since C is open, the set $U = C - x_0$ is an open neighborhood of the origin. However, to use the standard Minkowski functional construction, we prefer a balanced neighborhood.

In a locally convex space, we can define the functional p relative to an open convex set V containing 0. Let $V = C - z$ for some fixed $z \in C$. But let's use the simpler shift:

Pick $v_0 \in C$. Let $W = C - v_0$. Now $0 \in W$ and W is open and convex.

Define the Minkowski functional $p : X \to \mathbb{R}$:

$$p(x) = \inf\{t > 0 : x/t \in W\}$$

Because W is an open convex neighborhood of 0 in a TVS:

$p(x + y) \leq p(x) + p(y)$

$p(\lambda x) = \lambda p(x)$ for $\lambda \geq 0$

Continuity: There exists a neighborhood of zero where $p(x) < 1$, which in a TVS implies p is continuous.

$W = \{x \in X : p(x) < 1\}$.

3. Linear Extension

Since $0 \notin C$, let $x_0 = -v_0$. Then $x_0 \notin W$ (if it were, $x_0 \in C - v_0 \implies 0 \in C$, a contradiction).

Therefore, $p(x_0) \geq 1$.

Define a linear functional g on the subspace $L = \text{span}\{x_0\}$ by:

$$g(tx_0) = t \cdot p(x_0)$$

Just as in the normed case, $g(x) \leq p(x)$ for all $x \in L$. By the Analytic Hahn-Banach Theorem, there exists a linear extension $f : X \to \mathbb{R}$ such that:

$$f(x) \leq p(x) \quad \forall x \in X$$

4. Continuity and Separation

Continuity: Since $f(x) \leq p(x)$ and p is continuous at the origin, f is bounded above on a neighborhood of zero, which makes the linear functional f continuous ($f \in X^*$).

Separation: For any $c \in C$, we have $c - v_0 \in W$.

Thus $p(c - v_0) < 1$.

$$f(c - v_0) \leq p(c - v_0) < 1$$

$$f(c) - f(v_0) < 1$$

Since $f(x_0) = p(x_0)$ and $x_0 = -v_0$, we have $-f(v_0) = p(x_0)$.

$$f(c) + p(x_0) < 1 \implies f(c) < 1 - p(x_0)$$

Since $p(x_0) \geq 1$, we have $f(c) < 0$ for all $c \in C$.

5. Final Step

Since $c = a - b$, we have $f(a - b) < 0 \implies f(a) < f(b)$ for all $a \in A, b \in B$.

This implies:

$$\sup_{a \in A} f(a) \leq \inf_{b \in B} f(b)$$

□

Step 1: Construction of a Disjoint Neighborhood. Since C is closed and $v \notin C$, the complement $V \setminus C$ is an open neighborhood of v. By the translation invariance of the topology, there exists an open neighborhood W of 0 such that $(v + W) \cap C = \emptyset$.

Step 2: Invoke Local Convexity. Because V is locally convex, there exists a convex, balanced, open neighborhood U of 0 such that $U \subset W$. Thus, $(v + U) \cap C = \emptyset$.

Step 3: The Minkowski Functional and Extension. Define $A = C - v$. Then A is closed, convex, and $U \cap A = \emptyset$. Since U is open and convex, we can apply the geometric form of the Hahn-Banach Theorem. There exists a continuous linear functional F and a constant s_0 such that:

$$F(x) \leq s_0 < F(u) \quad \forall x \in A, \forall u \in U$$

Since $0 \in U$, we have $F(0) = 0$, which implies $s_0 < 0$. Let $s_0 = -\alpha$ for some $\alpha > 0$.

Step 4: Final Translation. Substituting $x = c - v$ into the inequality:

$$F(c - v) \leq -\alpha$$

$$F(c) - F(v) \leq -\alpha \implies F(c) \leq F(v) - \alpha$$

Setting $s = F(v) - \alpha$, we obtain:

$$F(c) \leq s < F(v)$$

for all $c \in C$. This proves the existence of a hyperplane strictly separating the point from the set.

□

Problem-2 Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and concave, and define T to be the set of all affine functions above f:

$T = \{\phi : \phi \geq f, \phi \; is \; affine\}$ Let $g(x) = inf_{\phi \in T}\{\phi(x)\}$. Prove that f = g.

`Author's solution-2`

Proof. To prove that $f = g$, we must show two inequalities: $g(x) \geq f(x)$ and $g(x) \leq f(x)$ for all $x \in \mathbb{R}$.

By the definition of the set T, every function $\phi \in T$ satisfies $\phi(x) \geq f(x)$ for all $x \in \mathbb{R}$. Taking the infimum over all such ϕ at a fixed x, we maintain the inequality:

$$g(x) = \inf_{\phi \in T} \phi(x) \geq f(x)$$

Thus, $g \geq f$. Left to show,the converse.

The core of the proof is to show that for any arbitrary point $x_0 \in \mathbb{R}$, there exists an affine function $\phi_0 \in T$ such that $\phi_0(x_0) = f(x_0)$.

Define the **hypgraph** of f as the set of points lying on or below the graph:

$$\text{hyp } f = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y \leq f(x)\}$$

Because f is concave, hyp f is a convex set. Since f is continuous, hyp f is a closed set in $\mathbb{R}^2$.

Consider the point $P = (x_0, f(x_0))$. Since P lies on the graph of f, it is on the boundary of the closed convex set hyp f. By the **Supporting Hyperplane Theorem** [Rockafellar 1970], there exists a non-zero vector $(a, b) \in \mathbb{R}^2$ and a constant $c \in \mathbb{R}$ such that:

1. $ax + by \leq c$ for all $(x, y) \in \text{hyp } f$
2. $ax_0 + bf(x_0) = c$

First, we note that b cannot be 0. If $b = 0$, then $ax \leq c$ for all $x \in \mathbb{R}$. Since $a \neq 0$, this linear inequality cannot hold for all x on the real line. Thus, $b \neq 0$.

Second, since $(x, y-1) \in \text{hyp } f$ whenever $(x, y) \in \text{hyp } f$, we have $ax + b(y-1) \leq c$. Substituting $ax + by \leq c$, we see that $-b \leq 0$, or $b > 0$.

Construction of the Affine Function:

Dividing the inequality $ax + by \leq c$ by b (since $b > 0$), we get:

$$y \leq -\frac{a}{b}x + \frac{c}{b}$$

Let $m = -a/b$ and $d = c/b$. Define the affine function $\phi_0(x) = mx + d$.

- For any x, the point $(x, f(x)) \in \text{hyp } f$, so $f(x) \leq \phi_0(x)$. This implies $\phi_0 \in T$.
- At x_0, we have $ax_0 + bf(x_0) = c$, which implies $f(x_0) = -\frac{a}{b}x_0 + \frac{c}{b} = \phi_0(x_0)$.

Since $g(x_0) = \inf_{\phi \in T} \phi(x_0)$ and we have found a specific $\phi_0 \in T$ such that $\phi_0(x_0) = f(x_0)$, it follows that:

$$g(x_0) \leq f(x_0)$$

Combined with $g(x_0) \geq f(x_0)$, we conclude $f(x_0) = g(x_0)$. Since x_0 was arbitrary, $f = g$ for all $x \in \mathbb{R}$. □

1.3. Section-III

Definition 1.12 (Monotone)**.** Let X and Y be posets. We say that $f : X \to Y$ is monotone if and only if $f(x) \geq f(x')$ whenever $x \geq x'$.

Definition 1.13 (Metric space)**.** A metric space is a set S together with a metric d on S. The metric d is a function that takes in two points in S and assigns a notion of distance between these two points. For example, the familiar real line $\mathbb{R}$ is a metric space with the metric $d(a, b) = |a - b|$. In order to qualify as a proper metric, d must have the following properties for all $a, b, c \in S$:

1. $d(a, b) = 0 \iff a = b$;
2. $d(a, b) \geq 0$;
3. $d(a, b) = d(b, a)$;
4. $d(a, c) + d(c, b) \geq d(a, b)$.

Definition 1.14 (1-Lipschitz)**.** Let X be a metric space with metric d. We say that $f : X \to \mathbb{R}$ is 1-Lipschitz if $\frac{|f(x)-f(x')|}{d(x,x')} \leq 1$ for all $x, x' \in X$.

All 1-Lipschitz functions are continuous.

Problem-3a Let $f : \mathbb{R} \to \mathbb{R}$ be a concave function that's monotone, 1-Lipschitz, and 0-increasing ($f(0) \geq 0$). Let T be the set of all affine functions above f that are monotone, 1-Lipschitz, and 0-increasing:

$$T = \{\varphi : \varphi \geq f, \varphi \text{ is monotone, 1-Lipschitz, 0-increasing}\}$$

Let $g(x) = \inf_{\varphi \in T}\{\varphi(x)\}$. Prove that $f = g$.

(Try to sketch the graph of a concave, monotone, 1-Lipschitz, 0-increasing function that is not the infimum of the set of all affine functions above it that share the latter three properties. Get a feel for why you can't sketch such a graph)

`Author's solution-3a`

Proof. To prove that $f = g$ for the given set T, we must demonstrate that $f(x) \leq g(x)$ and $g(x) \leq f(x)$ for all $x \in \mathbb{R}$.

Proof

1. Establishing $g(x) \geq f(x)$

By the definition of the set T:

$$T = \{\varphi : \varphi \geq f, \varphi \text{ is monotone, 1-Lipschitz, 0-increasing}\}$$

Since every $\varphi \in T$ satisfies $\varphi(x) \geq f(x)$ for all x, the pointwise infimum must also satisfy this bound:

$$g(x) = \inf_{\varphi \in T} \varphi(x) \geq f(x)$$

2. Establishing $g(x) \leq f(x)$

To prove the reverse inequality, we show that for any $x_0 \in \mathbb{R}$, there exists an affine function $\phi_0(x) = mx + c$ such that $\phi_0 \in T$ and $\phi_0(x_0) = f(x_0)$.

Step A: Existence of a supporting line

Since f is concave and continuous, there exists a supporting line at x_0. That is, there exists an affine function $\phi_0(x) = mx + c$ such that:

$\phi_0(x) \geq f(x)$ for all $x \in \mathbb{R}$

$\phi_0(x_0) = f(x_0)$

Step B: Verification of Properties for $\phi_0 \in T$

We now verify that this specific ϕ_0 inherits the three constraints from f:

1. Monotonicity ($m \geq 0$):

   Suppose $m < 0$. Since f is monotone, for $x > x_0$, $f(x) \geq f(x_0)$. however, for the affine function, if $m < 0$, then as $x \to \infty$, $\phi_0(x) \to -\infty$. This would eventually result in $\phi_0(x) < f(x)$, contradicting the fact that ϕ_0 is a supporting line. Thus, $m \geq 0$.

2. 1-Lipschitz ($m \leq 1$):

   For a concave function, the slope m of a supporting line at x_0 must lie between the left and right derivatives: $f'_+(x_0) \leq m \leq f'_-(x_0)$. Since f is 1-Lipschitz, we have $|f(x) - f(y)| \leq |x - y|$, which implies that $|f'(x)| \leq 1$ wherever the derivative exists. Consequently, the slope m must satisfy $|m| \leq 1$. Combined with $m \geq 0$, we have $0 \leq m \leq 1$.

3. $\phi_0(0) \geq 0$:

   We are given $f(0) \geq 0$. Since ϕ_0 is a supporting line such that $\phi_0(x) \geq f(x)$ for all x, it follows that:

$$\phi_0(0) \geq f(0) \geq 0$$

Since ϕ_0 satisfies all conditions, $\phi_0 \in T$. Because $g(x_0)$ is the infimum of all such functions at x_0:

$$g(x_0) \leq \phi_0(x_0) = f(x_0)$$

Since x_0 was arbitrary, $g(x) \leq f(x)$ for all x. Combined with Step 1, we conclude $f = g$.

□

Problem -3b (converse of 3a) Let $\{\varphi_i\}_{i \in I}$ be a collection of affine functions $\mathbb{R} \to \mathbb{R}$ that are monotone, 1-Lipschitz, and 0-increasing. Let $f(x) = \inf_{i \in I} \varphi_i(x)$. Prove that f is continuous, concave, monotone, 1-Lipschitz, and 0-increasing.

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Author's solution-3b
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Proof. Let $\varphi_i(x) = m_i x + c_i$ for each $i \in I$. We verify each property of f systematically:

1. **Concavity:** By definition, the pointwise infimum of a collection of affine functions is concave. Let $x, x' \in \mathbb{R}$ and $p \in [0,1]$. For every $i \in I$:

$$\varphi_i(px + (1-p)x') = p\varphi_i(x) + (1-p)\varphi_i(x')$$

Since $\varphi_i(x) \geq f(x)$ and $\varphi_i(x') \geq f(x')$ for all i, we have:

$$\varphi_i(px + (1-p)x') \geq pf(x) + (1-p)f(x')$$

Taking the infimum over $i \in I$ on the left side yields:

$$f(px + (1-p)x') \geq pf(x) + (1-p)f(x')$$

Thus, f is concave.

2. **Monotonicity:** Let $x \geq x'$. Since each φ_i is monotone, $\varphi_i(x) \geq \varphi_i(x')$ for all $i \in I$. By the property of infima, since $\varphi_i(x') \geq f(x')$, we have:

$$\varphi_i(x) \geq f(x') \quad \forall i \in I$$

Taking the infimum over i on the left:

$$f(x) = \inf_{i \in I} \varphi_i(x) \geq f(x')$$

Thus, f is monotone.

3. **1-Lipschitz Continuity:** For each φ_i, the 1-Lipschitz property implies $\varphi_i(x) - \varphi_i(x') \leq |x - x'|$. This can be rewritten as:

$$\varphi_i(x) \leq \varphi_i(x') + |x - x'|$$

Taking the infimum over i on both sides:

$$\inf_{i \in I} \varphi_i(x) \leq \inf_{i \in I}(\varphi_i(x') + |x - x'|) = \left(\inf_{i \in I} \varphi_i(x')\right) + |x - x'|$$

Which gives $f(x) \leq f(x') + |x - x'|$, or $f(x) - f(x') \leq |x - x'|$. Swapping x and x' gives the reverse inequality, leading to:

$$|f(x) - f(x')| \leq |x - x'|$$

This proves f is 1-Lipschitz, which also implies that f is **continuous**.

4. **0-increasing:** Since each φ_i is 0-increasing, we have $\varphi_i(0) \geq 0$ for all $i \in I$. The infimum of a set of non-negative numbers is non-negative:

$$f(0) = \inf_{i \in I} \varphi_i(0) \geq 0$$

Thus, f is 0-increasing.

□

1.4. Section-IV

Definition 1.15 ($C(X)$)**.** Let X be a compact (definition in appendix) metric space. Then $C(X)$ is the vector space of all continuous functions $X \to \mathbb{R}$. Take some time to wrap your head around this vector space. Each vector in $C(X)$ is a function! Note that $C(X)$ comes equipped with the following metric:

$$d(f, g) = \max_{x \in X} |f(x) - g(x)|.$$

Also, every function in $C(X)$ is bounded above and below (a consequence of X being compact).

Problem 4a Let $\psi : C(X) \to \mathbb{R}$ (note that this type signature makes ψ a functional) be concave, monotone, 1-Lipschitz, and 0-increasing. Prove that

$$\psi = \inf\{\varphi : C(X) \to \mathbb{R} : \varphi \geq \psi, \varphi \text{ affine, mono, 1-Lipschitz, and } \varphi(0) \geq 0\}.$$

```
Author's solution-4a
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Proof. Let $V = C(X)$ be the space of continuous functions on X, and let Ψ be the set of affine functionals $\varphi : V \to \mathbb{R}$ that are monotone, 1-Lipschitz, and satisfy $\varphi(0) \geq 0$. We define:

$$g(f) = \inf_{\varphi \in \Psi} \varphi(f)$$

We want to show $g = \psi$.

1. The Direction $g \geq \psi$

By the definition of the set Ψ, every $\varphi \in \Psi$ satisfies $\varphi(f) \geq \psi(f)$ for all $f \in C(X)$. Taking the infimum over Ψ maintains this bound:

$$g(f) = \inf_{\varphi \in \Psi} \varphi(f) \geq \psi(f)$$

2. The Direction $g \leq \psi$

We need to show that for any $f_0 \in C(X)$, there exists an affine functional $\varphi_0 \in \Psi$ such that $\varphi_0(f_0) = \psi(f_0)$.

Step A: Finding a Supporting Hyperplane

Define the hypograph of ψ in $C(X) \times \mathbb{R}$:

$$\text{hyp } \psi = \{(f, y) \in C(X) \times \mathbb{R} \mid y \leq \psi(f)\}$$

Since ψ is concave and continuous (1-Lipschitz implies continuity), hyp ψ is a closed convex set. The point $(f_0, \psi(f_0))$ is on its boundary. By the Hahn-Banach Separation Theorem (specifically the supporting hyperplane version [Lax 2002] for the interior of a convex set), there exists a continuous linear functional $L : C(X) \to \mathbb{R}$ such that the affine functional $\varphi_0(f) = L(f - f_0) + \psi(f_0)$ satisfies:

$$\varphi_0(f) \geq \psi(f) \quad \forall f \in C(X)$$

By construction, $\varphi_0(f_0) = \psi(f_0)$.

Step B: Verifying $\varphi_0 \in \Psi$

We must verify that φ_0 (which is affine by construction) inherits the required properties:

1-Lipschitz: Since $\varphi_0 \geq \psi$ and $\varphi_0(f_0) = \psi(f_0)$, then for any $f \in C(X)$ and $\epsilon > 0$, $\psi(f_0 + \epsilon f) \leq \varphi_0(f_0 + \epsilon f)$. Using the 1-Lipschitz property of ψ:

$$\psi(f_0 + \epsilon f) - \psi(f_0) \leq \|\epsilon f\| = \epsilon \|f\|$$

Dividing by ϵ and taking the limit as $\epsilon \to 0$ (noting that for affine functionals, the directional derivative is the linear part L):

$$L(f) \leq \|f\|$$

This implies the norm of the linear functional $\|L\| \leq 1$, making φ_0 1-Lipschitz.

Monotonicity: Suppose $f \geq 0$. If $L(f) < 0$, then for large $t > 0$, $\varphi_0(f_0 + tf) = \psi(f_0) + tL(f)$ would eventually be smaller than $\psi(f_0 + tf)$ because ψ is monotone increasing. This contradicts $\varphi_0 \geq \psi$. Thus, L must be a positive linear functional, so φ_0 is monotone.

0-increasing: Since $\varphi_0 \geq \psi$ everywhere and $\psi(0) \geq 0$, we have:

$$\varphi_0(0) \geq \psi(0) \geq 0$$

Now,

Since $\varphi_0 \in \Psi$ and $\varphi_0(f_0) = \psi(f_0)$, the infimum $g(f_0)$ cannot be greater than $\psi(f_0)$. Thus $g \leq \psi$, concluding the proof that $\psi = g$. □

Problem 4b Let $\{\varphi_i\}_{i \in I}$ be a collection of affine functions $C(X) \to \mathbb{R}$ that are monotone, 1-Lipschitz, and 0-increasing. Let $f(x) = \inf_{i \in I} \varphi_i(x)$. Prove that f is continuous, concave, monotone, 1-Lipschitz, and 0-increasing.

This proof is analogous to the proof of Problem 3b, but do go through the exercise for the practice.

```
Author's solution- 4b
```

Proof. Let $V = C(X)$ be the space of continuous functions on a compact metric space X, equipped with the supremum norm $\|f\| = \max_{x \in X} |f(x)|$. Let $f(h) = \inf_{i \in I} \varphi_i(h)$ for $h \in C(X)$.1. ConcavityLet $h, k \in C(X)$ and $p \in [0, 1]$. Since each φ_i is affine (and thus concave):

$$\varphi_i(ph + (1-p)k) = p\varphi_i(h) + (1-p)\varphi_i(k)$$

By the property of the infimum, $\varphi_i(h) \geq f(h)$ and $\varphi_i(k) \geq f(k)$. Substituting these:

$$\varphi_i(ph + (1-p)k) \geq pf(h) + (1-p)f(k)$$

Taking the infimum over $i \in I$ on the left side:

$$f(ph + (1-p)k) = \inf_{i \in I} \varphi_i(ph + (1-p)k) \geq pf(h) + (1-p)f(k)$$

Thus, f is concave.2. MonotonicityLet $h, k \in C(X)$ such that $h \geq k$ (meaning $h(x) \geq k(x)$ for all $x \in X$).Since each φ_i is monotone, $\varphi_i(h) \geq \varphi_i(k)$.By the definition of the infimum:

$$\varphi_i(h) \geq \varphi_i(k) \geq \inf_{j \in I} \varphi_j(k) = f(k)$$

Since $\varphi_i(h) \geq f(k)$ for all $i \in I$, it follows that:

$$f(h) = \inf_{i \in I} \varphi_i(h) \geq f(k)$$

Thus, f is monotone.3. 1-Lipschitz ContinuityEach φ_i is 1-Lipschitz with respect to the supremum norm: $|\varphi_i(h) - \varphi_i(k)| \leq \|h - k\|$.This implies:

$$\varphi_i(h) \leq \varphi_i(k) + \|h - k\|$$

Taking the infimum over $i \in I$ on both sides:

$$\inf_{i \in I} \varphi_i(h) \leq \inf_{i \in I}(\varphi_i(k) + \|h - k\|) = \left(\inf_{i \in I} \varphi_i(k)\right) + \|h - k\|$$

So, $f(h) \leq f(k) + \|h - k\|$, which implies $f(h) - f(k) \leq \|h - k\|$. By symmetry:

$$|f(h) - f(k)| \leq \|h - k\|$$

This proves f is 1-Lipschitz. Since any Lipschitz function is continuous, f is continuous.4. 0-increasingWe are given that $\varphi_i(0) \geq 0$ for all $i \in I$, where 0 represents the zero function in $C(X)$.The infimum of a set of non-negative values is non-negative:

$$f(0) = \inf_{i \in I} \varphi_i(0) \geq 0$$

Thus, f is 0-increasing.

□

Discussion: Our work here allows us to define the fundamental object of Infra-Bayesianism from two different perspectives:

Definition 1.16 (Inframeasure (Functional View))**.** An inframeasure is any functional $\psi : C(X) \to \mathbb{R}$ that has the following properties:

1. Concave
2. Monotone
3. 1-Lipschitz
4. 0-increasing.

Definition 1.17 (Inframeasure(set view))**.** An inframeasure is a set $\{\varphi_i\}_{i \in I}$ of functionals $C(X) \to \mathbb{R}$ that have the following properties:

1. Affine
2. Monotone
3. 1-Lipschitz
4. 0-increasing.

Theorem 1.18 (LF-Duality). *[Rockafellar 1970] A shorthand for saying that one can switch back and forth freely be- tween the functional view and set view of inframeasures is to say that*

inframeasures have LF-Duality. This is short for Legendre-Fenchel duality, because the same concept applied in another branch of math gives rise to something called the Legendre-Fenchel transform.

We just proved the Fundamental Theorem of Inframeasures! The big theorem that gets used literally all the time, and you mastered all the fiddly details of it.

What this proved is that we can toggle back and forth between a concave function $\psi : C(X) \to \mathbb{R}$ and the set of affine functions above it. In order to really get anywhere with it, we need to figure out what the affine functions mean. We'll take a brief detour for the next few problems and then come back to restate the full-power version of the Fundamental Theorem of Inframeasures.

From the first thing you proved, any affine function $V \to \mathbb{R}$ can be split into a constant and a linear function. We'll now be covering what the linear function means.

1.5. section-V

Problem 5a Come up with a concrete example of a linear functional $C(X) \to \mathbb{R}$. Think simple. Just think of X as $[0, 1]$. What is the classic function(al) that takes a function as input and returns a real number?

`Author's solution-5a`

Proof. To prove that L is a linear functional, we must satisfy two conditions for all $f, g \in C([0, 1])$ and all scalars $\alpha \in \mathbb{R}$.1. AdditivityWe examine the functional applied to the sum of two functions:

$$L(f + g) = \int_0^1 (f(x) + g(x))\, dx$$

By the linearity property of the Riemann integral (the integral of a sum is the sum of the integrals):

$$\int_0^1 (f(x) + g(x))\, dx = \int_0^1 f(x)\, dx + \int_0^1 g(x)\, dx$$

Thus:

$$L(f + g) = L(f) + L(g)$$

2. Homogeneity (Scalar Multiplication)We examine the functional applied to a scaled function:

$$L(\alpha f) = \int_0^1 (\alpha f(x))\, dx$$

By the constant multiple rule for integration:

$$\int_0^1 \alpha f(x)\, dx = \alpha \int_0^1 f(x)\, dx$$

Thus:

$$L(\alpha f) = \alpha L(f)$$

Since L satisfies both additivity and homogeneity, L is a linear functional on $C([0, 1])$.

Additional Properties (Relevance to Inframeasures) : it is worth noting that this specific functional L also satisfies:

Monotonicity: If $f(x) \geq g(x)$ for all x, then $\int f \geq \int g$.

1-Lipschitzness: $|L(f) - L(g)| = |\int (f - g)| \leq \int |f - g| \leq \max |f - g| \cdot (1 - 0) = d(f, g)$

Normalization: $L(1) = 1$. □

Definition 1.19 (Measure)**.** Let X be a set. A measure is a function m that takes in subsets of X (technically, m can't take in any old subset of X; the subset has to be in the σ-algebra of X, but don't worry too much about that) and maps these subsets of X to a real number corresponding to the "size" of the subset. A measure must have the following properties: 1) $m(\emptyset) = 0$; 2) Countable additivity: For all countable collections $\{E_k\}_{k=1}^{\infty}$ of pair- wise disjoint subsets of X,

$$m\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} m(E_k);$$

3) For all $E \subseteq X$, $m(E) \geq 0$. If $m(X) = 1$, we say that m is a probability measure (aka probability distribution).

Definition 1.20 (Finite Signed Measure)**.** If $m : X \to \mathbb{R}$ meets the first two requirements but not the third, and if $m(E)$ is finite $\forall E \subseteq X$, we say that m is a finite signed measure.

Any finite signed measure can be uniquely decomposed into two mea- sures m^+ and m^- (both of which meet the third requirement) such that $m(E) = m^+(E) - m^-(E)$. It's exactly the same as how you can decompose any function $f : X \to \mathbb{R}$ into a positive and negative part: $f = \max(f, 0) - \max(-f, 0)$ where when one of these parts is positive, the other must be 0. The space of all finite signed measures on X is a vector space, denoted $M^{\pm}(X)$.

Definition 1.21 (Expectation)**.** The term "expectation" conflates two importantly different concepts. The first corresponds to a weighted average: Suppose that we have a sample space Ω and a continuous random variable Y on Ω with prob- ability density function $f(y)$. In introductory probability courses, the expected value of Y is defined as follows:

$$E(Y) = \int_{\mathbb{R}} y f(y) dy$$

We can call this operation averaging to distinguish it from this sec- ond concept: Given probability space (X, Σ, m), the expectation of a continuous random variable g on X is given by

$$E(g) = \int_X g(x) dm$$

Problem 5b Let $m \in M^{\pm}(X)$. Prove that $\psi(g) = \int_X g(x)dm$ is a continuous linear functional $C(X) \to \mathbb{R}$. In other words, prove that taking expectations of functions with respect to a finite signed measure is a continuous linear mapping. Tip: For this proof, you can assume that $\forall m \in M^{\pm}(X)$,

$$\int_X f(x) d(m^+ - m^-) = \int_X f(x) dm^+ - \int_X f(x) dm^-$$

`Author's solution-5b`

Proof. Let $\psi : C(X) \to \mathbb{R}$ be defined by $\psi(g) = \int_X g\, dm$.

1. **Linearity:** For any $g, h \in C(X)$ and $\alpha \in \mathbb{R}$, the linearity of the integral over a signed measure implies:

$$\psi(\alpha g + h) = \int_X (\alpha g + h)\, dm = \alpha \int_X g\, dm + \int_X h\, dm = \alpha\psi(g) + \psi(h)$$

   Thus, ψ is linear.

2. **Continuity:** A linear functional is continuous if and only if it is bounded. We use the Jordan decomposition [Folland 1999],

   $m = m^+ - m^-$. For any $g \in C(X)$:

$$|\psi(g)| = \left| \int_X g\, dm^+ - \int_X g\, dm^- \right| \leq \left| \int_X g\, dm^+ \right| + \left| \int_X g\, dm^- \right|$$

   .

Using the property $|\int g\, d\mu| \leq \|g\|_\infty \mu(X)$ for positive measures:

$$|\psi(g)| \leq \|g\|_\infty m^+(X) + \|g\|_\infty m^-(X) = \|g\|_\infty (m^+(X) + m^-(X))$$

.

Since m is a finite signed measure, the total variation $\|m\| = m^+(X) + m^-(X)$ is a finite real number. Therefore, $|\psi(g)| \leq \|m\| \cdot \|g\|_\infty$, which proves ψ is bounded and thus continuous.

□

Problem-5c Come up with a continuous linear functional $F : C(X) \to \mathbb{R}$ that does not involve integrating with respect to some $m \in M^{\pm}(X)$. In other words, find a continuous linear functional that does not look like this:

$$F(g) = \int_X g(x)dm$$

`Author's solution-5c`

Proof. Let X be a compact metric space. We define the evaluation functional $F : C(X) \to \mathbb{R}$ at a fixed point $x_0 \in X$ as:

$$F(g) = g(x_0)$$

1. **Linearity:** For any $g, h \in C(X)$ and $\alpha, \beta \in \mathbb{R}$, we have:

$$F(\alpha g + \beta h) = (\alpha g + \beta h)(x_0) = \alpha g(x_0) + \beta h(x_0) = \alpha F(g) + \beta F(h)$$

   This satisfies the requirements for a linear mapping.

2. **Continuity:** A linear functional on a normed space is continuous if it is bounded. The norm on $C(X)$ is the supremum norm $\|g\| = \sup_{x \in X} |g(x)|$. For any function g:

$$|F(g)| = |g(x_0)| \leq \sup_{x \in X} |g(x)| = \|g\|$$

   Since $|F(g)| \leq 1 \cdot \|g\|$, the functional is bounded and thus continuous.

While this functional does not look like a standard integral (e.g., a Riemann integral), it is formally equivalent to integration with respect to the Dirac measure δ_{x_0}:

$$F(g) = \int_X g(x)\, d\delta_{x_0}$$

The Riesz-Markov-Kakutani theorem [Rudin 1991] guarantees that *all* continuous linear functionals on $C(X)$ are represented by some finite signed measure $m \in M^{\pm}(X)$. □

That's right, no such function F exists! Every single continuous linear func- tional $C(X) \to \mathbb{R}$ is an integral with respect to some finite signed measure. No need to prove this. Fun fact: It's called the Riesz-Markov-Kakutani Represen- tation Theorem.

We've just shown something very interesting in the previous two problems: Fi- nite signed measures on X equal continuous linear functionals $C(X) \to \mathbb{R}$. The notion that "the important part of measures is that you can take expectations with them" is even more true than it looked like. The thing you do with mea- sures is take expectations of functions with them. If you want to find the prob- ability of an event, just take the expectation of the function that's 1 where the event happens and 0 everywhere else. Measures literally are continuous linear functionals that take functions in and give expectations out. And conversely, every continuous linear way of taking functions in and returning expectations corresponds to some finite signed measure. Thus, from now on, we go with notation that emphasizes this perspective of how measures take functions as input: Writing $m(g)$ is the same as writing $\int_X g(x)dm$, which is the same as writing $E_{x \sim m}[g(x)]$.

Definition 1.22 (Indicator Function)**.** Let A be a set. Then the function $\mathbf{1}_A$ (the $\mathbf{1}$ is in bold to show that it's a function and to distinguish this object from the number 1) is called an indicator function. For all $a \in A$, $\mathbf{1}_A(a) = 1$, and for all $b \notin A$, $\mathbf{1}_A(b) = 0$. Note that if X is the compact metric space we've been dealing with for the past several problems then

$$m(\mathbf{1}_X) = \int_X \mathbf{1}_X dm = m(X)$$

On the far left of this equation, we're viewing m as a linear functional that takes functions as inputs, and on the far right, we're viewing m as a measure that takes in sets.

Problem - 5d What extra conditions on a continuous linear functional must be added to get it to correspond to a measure m that isn't a signed measure? In other words, how can we make it so that $m(E) \geq 0$ for all $E \subseteq X$?

What additional conditions must be imposed on a continuous linear func- tional to get it to correspond to a probability distribution? ("Probability measure" and "probability distribution" mean the same thing.)

```
Author's solution - 5d
```

Proof. Let $m : C(X) \to \mathbb{R}$ be a continuous linear functional.

1. **Non-negativity:** By the Riesz Representation Theorem, there exists a unique signed measure μ such that $m(f) = \int_X f\, d\mu$. If m is monotone ($f \geq 0 \implies m(f) \geq 0$), then for any measurable set E, we can approximate the indicator $\mathbf{1}_E$ with a sequence of non-negative continuous functions f_n. By the Monotone Convergence Theorem, $\mu(E) = \lim \int f_n\, d\mu \geq 0$.

2. **Probability Measure:** If in addition to monotonicity, we require $m(\mathbf{1}_X) = 1$, then:

$$\mu(X) = \int_X 1\, d\mu = m(1) = 1$$

A non-negative measure with total mass 1 is, by definition, a probability distribution.

□

1.6. section-VI

Definition 1.23 (A-Measure)**.** An a-measure (short for affine measure) is a pair (m, b) of a finite measure m and a number $b \geq 0$.

An sa-measure (short for signed a-measure) is exactly the same as an a-measure except that m is a finite signed measure. This definition won't be used in this problem set, but we're introducing it here because it comes up often in the Alignment Forum posts about infra-Bayesianism.

One thing to note is that we'll need a special term for referring to the set of sa-measures where $b \geq m^-(1)$, because this is a particularly relevant subset.

Problem 6 Let $\varphi : C(X) \to \mathbb{R}$ be affine, monotone, 1-Lipschitz, and 0-increasing. Prove that φ corresponds to an a-measure (m, b) where $0 \leq m(X) \leq 1$.

```
Author's solution-6
```

Proof. Let $\varphi : C(X) \to \mathbb{R}$ be an affine, monotone, 1-Lipschitz, and 0-increasing functional.1. Splitting the Affine FunctionalFrom Problem 1, we know that any affine functional φ can be decomposed into a linear part and a constant part:

$$\varphi(g) = L(g) + c$$

where $L : C(X) \to \mathbb{R}$ is a linear functional and $c = \varphi(0)$.2. Identifying the Measure m and the Bias bBy the Riesz-Markov-Kakutani Representation Theorem, every continuous linear functional L on $C(X)$ corresponds to a unique finite signed measure m such that $L(g) = \int_X g\, dm$.Thus, we can write:

$$\varphi(g) = \int_X g\, dm + c$$

We define the a-measure as the pair (m, c). We now verify the constraints on m and c (which the definition calls b).3. Proving m is a Finite (Non-negative) MeasureThe functional φ is given as monotone. For any $g, h \in C(X)$ with $g \geq h$:

$$\varphi(g) \geq \varphi(h) \implies L(g) + c \geq L(h) + c \implies L(g) \geq L(h)$$

This implies the linear functional L is monotone. As established in Problem 5d, a monotone linear functional corresponds to a non-negative measure. Thus, m is a finite measure ($m(E) \geq 0$ for all E).4. Proving $b \geq 0$ (0-increasing)The functional is 0-increasing, meaning $\varphi(0) \geq 0$. Since $\varphi(0) = L(0) + c = 0 + c$, we have:

$$c \geq 0$$

Letting $b = c$, we have $b \geq 0.5$. Proving $0 \leq m(X) \leq 1$ (1-Lipschitz)The functional φ is 1-Lipschitz. For any $g, h \in C(X)$:

$$|\varphi(g) - \varphi(h)| \leq \|g - h\|$$

Substituting the decomposition:

$$|(L(g) + b) - (L(h) + b)| = |L(g) - L(h)| = |L(g - h)| \leq \|g - h\|$$

This shows that the operator norm of the linear functional L satisfies $\|L\| \leq 1$.For a non-negative measure m, the norm $\|L\|$ is attained by the constant function $\mathbf{1}_X$:

$$\|L\| = L(\mathbf{1}_X) = \int_X \mathbf{1}_X \, dm = m(X)$$

Therefore, $m(X) \leq 1$. Since m is a non-negative measure, we also have $m(X) \geq 0$. □

Discussion: In Sections 1-4, we defined inframeasures and showed that they have LF-Duality. That is, we can switch seamlessly back and forth between the set view and functional view of inframeasures (go back to Section 4 and read the definition of an inframeasure if you've forgotten.) In Sections 5 and 6, we learned the following three things:

1) "Finite signed measure," "continuous linear functional $C(X) \to \mathbb{R}$," and "a way to take expectations of functions in $C(X)$ in a continuous, linear way" are all identical ways to refer to the same object.

2) If m is a measure that has no negative part, then m corresponds to a mono- tone linear functional. Probability distributions correspond to linear functional that are monotone and 1-Lipschitz.

3) Affine, monotone, 1-Lipschitz, 0-increasing functionals can be thought of as (m, b) pairs, where $0 \leq m(X) \leq 1$ and $b \geq 0$.

With these ideas under our belts, it's time to return to the LF-duality dis- cussion and get a beefed-up version of the Fundamental Theorem of Infra- Measures.

1.7. Section-VII

The Fundamental Theorem of Inframeasures is just a restatement of claims 4a and 4b using the new concepts we just went over.

Theorem 1.24 (Fundamental Theorem of Inframeasures (Forwards Direction))**.** *Let $\psi : C(X) \to \mathbb{R}$ be concave, monotone, 1-Lipschitz, and 0-increasing. Let T be the set of all a-measures above ψ.*

$$T = \{(m, b) : \forall g \in C(X), m(g) + b \geq \psi(g)\}$$

Define ψ^* as follows:

$$\psi^*(g) = \inf_{(m,b) \in T} \{m(g) + b\}$$

Prove that $\psi = \psi^*$.

`Author's solution`

Proof. Let $\psi : C(X) \to \mathbb{R}$ be concave, monotone, 1-Lipschitz, and 0-increasing. We wish to show that $\psi(g) = \psi^*(g)$ for all $g \in C(X)$.1. Establishing $\psi^* \geq \psi$By the definition of the set T:

$$T = \{(m, b) : m(g) + b \geq \psi(g) \text{ for all } g \in C(X)\}$$

For any specific a-measure $(m, b) \in T$, the inequality $m(g) + b \geq \psi(g)$ holds by construction. Taking the infimum over all such pairs (m, b) in T preserves this inequality:

$$\psi^*(g) = \inf_{(m,b) \in T} \{m(g) + b\} \geq \psi(g)$$

2. Establishing $\psi^* \leq \psi$To prove the reverse, we must show that for any arbitrary $g_0 \in C(X)$, there exists a specific a-measure $(m_0, b_0) \in T$ such that $m_0(g_0) + b_0 = \psi(g_0)$.Step A: Finding a Supporting Affine

FunctionalSince ψ is a concave and continuous (1-Lipschitz) functional on the Banach space $C(X)$, we apply the Supporting Hyperplane Theorem [Lax 2002].

For any g_0, there exists a continuous linear functional $L : C(X) \to \mathbb{R}$ such that:

$$L(g - g_0) + \psi(g_0) \geq \psi(g) \quad \forall g \in C(X)$$

We can rewrite this as:

$$L(g) + \underbrace{(\psi(g_0) - L(g_0))}_{b_0} \geq \psi(g)$$

Let $\varphi(g) = L(g) + b_0$. By construction, $\varphi(g_0) = \psi(g_0)$ and $\varphi \geq \psi$.Step B: Verifying φ corresponds to an a-measure in TFrom the result in Problem 6, we know that since φ is an affine functional above ψ, and ψ is monotone, 1-Lipschitz, and 0-increasing, φ inherits these properties:Monotonicity: Since $\varphi \geq \psi$ and ψ is monotone, L must be a positive linear functional, corresponding to a non-negative measure m_0.1-Lipschitz: Since $\varphi \geq \psi$ and they touch at g_0, the slope L must satisfy $\|L\| \leq 1$, meaning $m_0(X) \leq 1$.0-increasing: Since $\varphi(0) \geq \psi(0) \geq 0$, the bias $b_0 \geq 0$.Thus, the pair (m_0, b_0) is an a-measure and $(m_0, b_0) \in T$.ConclusionBecause $(m_0, b_0) \in T$ and $m_0(g_0) + b_0 = \psi(g_0)$, it follows that:

$$\psi^*(g_0) = \inf_{(m,b)\in T} \{m(g_0) + b\} \leq m_0(g_0) + b_0 = \psi(g_0)$$

Since we have $\psi^* \geq \psi$ and $\psi^* \leq \psi$, we conclude $\psi = \psi^*$. □

Theorem 1.25 (Fundamental Theorem of Inframeasures (Backwards Direction))**.** *For all sets Ψ of a-measures with $0 \leq m(1) \leq 1$, prove that the following function is concave, monotone, 1-Lipschitz, and 0-increasing:*

$$\lambda(g) = \inf_{(m,b)\in\Psi} \{m(g) + b\}$$

`Author's solution`

Proof. Let Ψ be a set of a-measures (m, b) such that $0 \leq m(\mathbf{1}_X) \leq 1$ and $b \geq 0$. Let $\lambda(g) = \inf_{(m,b)\in\Psi}\{m(g) + b\}$.

1. Concavity

For each $(m, b) \in \Psi$, the functional $\varphi(g) = m(g) + b$ is affine. Let $g, h \in C(X)$ and $p \in [0, 1]$.

$$\varphi(pg + (1 - p)h) = p\varphi(g) + (1 - p)\varphi(h)$$

Since $\varphi(g) \geq \lambda(g)$ and $\varphi(h) \geq \lambda(h)$ for all $\varphi \in \Psi$:

$$\varphi(pg + (1 - p)h) \geq p\lambda(g) + (1 - p)\lambda(h)$$

Taking the infimum over all $(m, b) \in \Psi$:

$$\lambda(pg + (1 - p)h) = \inf \varphi(pg + (1 - p)h) \geq p\lambda(g) + (1 - p)\lambda(h)$$

Thus, λ is concave.

2. Monotonicity

Let $g \geq h$ in $C(X)$. For each a-measure, m is a non-negative measure, so the linear part $m(\cdot)$ is monotone.

$$m(g) + b \geq m(h) + b \geq \inf_{(m',b')\in\Psi} \{m'(h) + b'\} = \lambda(h)$$

Since $m(g) + b \geq \lambda(h)$ for every $(m, b) \in \Psi$:

$$\lambda(g) = \inf_{(m,b)\in\Psi} \{m(g) + b\} \geq \lambda(h)$$

Thus, λ is monotone.

3. 1-Lipschitz

Each affine functional $\varphi(g) = m(g) + b$ is 1-Lipschitz because the norm of the linear functional m is $\|m\| = m(\mathbf{1}_X) \leq 1$. Therefore:

$$\varphi(g) \leq \varphi(h) + \|g - h\|$$

Taking the infimum over $(m, b) \in \Psi$ on both sides:

$$\inf \varphi(g) \leq \inf(\varphi(h) + \|g - h\|) = (\inf \varphi(h)) + \|g - h\|$$

This gives $\lambda(g) - \lambda(h) \leq \|g - h\|$. By symmetry, $|\lambda(g) - \lambda(h)| \leq \|g - h\|$.
Thus, λ is 1-Lipschitz.
4. 0-increasing
For each $(m, b) \in \Psi$, we have $b \geq 0$ and m is non-negative, so $m(\mathbf{0}) = 0$.

$$\varphi(\mathbf{0}) = m(\mathbf{0}) + b = b \geq 0$$

Since every functional in the set is non-negative at $\mathbf{0}$, their infimum must be non-negative:

$$\lambda(\mathbf{0}) = \inf_{(m,b)\in\Psi} \varphi(\mathbf{0}) \geq 0$$

Thus, λ is 0-increasing. □

Discussion: We can now modify definition 4.2b:

> **Definition 1.26** (Inframeasure (Set View))**.** An inframeasure is what you get when you do worst case selection over a set of a-measures. In the special case where each a-measure in the set takes the form $(m, 0)$ where m is a probability distribution, an inframeasure is the result of worst-case selection over a set of probability distributions. We call this special type of inframeasure an infradistribution.
>
> Note: This terminology has evolved slightly over time. In all of Diffractor's previous posts, he used the term "crisp infradistribution" to refer to the concept we're defining here and used "infradistribution" to define a different concept. Now, he is switching to using the term "infradistribu- tion" in the way we have defined it here (to refer to what he previously called a crisp infradistribution).
>
> You may be wondering, "Why do we need to build up the theory of general inframeasures? Aren't infradistributions the thing we really care about if we want to apply all of this math to the theory of reinforcement learning?"
>
> Great question. There are several reasons why we need the general theory of inframeasures. One of the main reasons is that eventually we will want to un- derstand how updating works for inframeasures (a future problem set much like this one will guide you through developing the updating theory for inframea- sures.) If we just deal with infradistributions, we can't formulate a dynamically consistent updating strategy. This is a major shortcoming of "imprecise proba- bility theory," a branch of math that Vanessa and Diffractor drew on in order to invent IB. Working with fully generalized inframeasures allows us to update in a dynamically consistent way.

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